

Model card doc, draft 1

This document will eventually become the model card, for the intrinsic value analyzer, but currently contains a draft with project details

Model calculation implementation guides:

- ▶ How to Calculate the Intrinsic Value of a Stock like Benjamin Graham! (Step by Step)
- ▶ How To Calculate Intrinsic Value (Full Example)

Data required for model:

Intrinsic value - calculated using benjamin graham's formula for intrinsic value of a stock

- EPS (earning per share)
- P/E ratio
- Growth Rate
- Average Yield of AAA bonds = 4.4 (constant)
- Current yield of AAA bonds = 5.3 (2026-03-16)

The training data should have the following values:

- Intrinsic value
- Share price
- Sector name

If a training dataset with the specific values required can't be found then the Intrinsic value can be calculated and added to the training data, all values required to calculate the intrinsic value must be present at minimum.

Model packaging:

```
# 1. Define your custom formula
def my_custom_formula(X):
    # Example: Calculate a ratio or interaction between columns
    combined_feature = X[:, 0] * X[:, 1]
    return np.column_stack((X, combined_feature))

# 2. Build the Pipeline
pipeline = Pipeline([
    ('custom_math', FunctionTransformer(my_custom_formula)),
    ('neural_net', Perceptron()) # Or any MLPClassifier
])

# 3. Save the whole thing
import joblib
joblib.dump(pipeline, 'full_model.joblib')
```

Model calculations:

- $\text{EPS (earning per share)} = \text{net income} / \text{number of shares}$
- $\text{P/E ratio} = \text{price per share} / \text{EPS}$
- $\text{Growth Rate} = (\text{avg. 5 year price growth} + \text{avg. 5 year return growth}) / 2$
- $\text{Current yield of AAA bonds} = \text{find online on the yahoo moody tracker}$

Model data sources:

[s&p 500 EPS and PE ratio](#)

[Current Corporate AAA bond yield](#)

[s&p 500 growth rate price/returns](#)